

Name: Monisha Raghini Chelle

RollNo: 2520030537

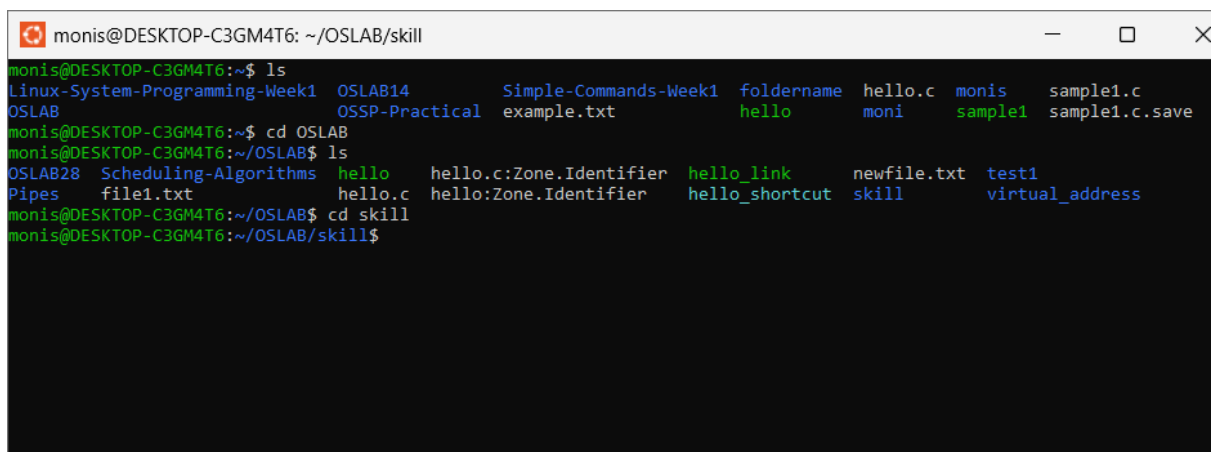
Section-6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

24. To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSLAB14      Simple-Commands-Week1  foldername  hello.c  monis  sample1.c
OSLAB                          OSSP-Practical  example.txt            hello       moni    sample1  sample1.c.save
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello      hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c    hello:Zone.Identifier    hello_shortcut  skill        virtual_address
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 1 — Generate the source-file list

Run these commands:

```
cd ~/OSLAB/skill
```

```
find ~/OSLAB/skill -name "*.c" | sort > source_files.txt
```

```
cat source_files.txt
```

```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ find ~/OSLAB/skill -name "*.c" | sort > source_files.txt
cat source_files.txt
/home/monis/OSLAB/skill/append_redirection.c
/home/monis/OSLAB/skill/background_jobs.c
/home/monis/OSLAB/skill/builtin_dispatch.c
/home/monis/OSLAB/skill/builtins.c
/home/monis/OSLAB/skill/child_process.c
/home/monis/OSLAB/skill/combined_redirection.c
/home/monis/OSLAB/skill/command_history.c
/home/monis/OSLAB/skill/command_launch.c
/home/monis/OSLAB/skill/complex_commands.c
/home/monis/OSLAB/skill/complex_commands1.c
/home/monis/OSLAB/skill/directory_navigation.c
/home/monis/OSLAB/skill/directory_state.c
/home/monis/OSLAB/skill/double_quotes.c
/home/monis/OSLAB/skill/error_handling.c
/home/monis/OSLAB/skill/escape_parser.c
/home/monis/OSLAB/skill/execution_error.c
/home/monis/OSLAB/skill/export_child.c
/home/monis/OSLAB/skill/export_variables.c
/home/monis/OSLAB/skill/foreground_switch.c
/home/monis/OSLAB/skill/history.c
/home/monis/OSLAB/skill/input_redirection.c
/home/monis/OSLAB/skill/integrated_shell.c
/home/monis/OSLAB/skill/job_table.c
/home/monis/OSLAB/skill/list_jobs.c
/home/monis/OSLAB/skill/long_pipeline.c
/home/monis/OSLAB/skill/memory.c
/home/monis/OSLAB/skill/memory1.c
/home/monis/OSLAB/skill/memory2.c
/home/monis/OSLAB/skill/memory3.c
/home/monis/OSLAB/skill/memory4.c
/home/monis/OSLAB/skill/memory_test.c
/home/monis/OSLAB/skill/multiple_pipes.c
/home/monis/OSLAB/skill/output_redirection.c
/home/monis/OSLAB/skill/parent_child.c
/home/monis/OSLAB/skill/parser_logic.c
/home/monis/OSLAB/skill/pass_arguments.c
/home/monis/OSLAB/skill/path_missing.c
/home/monis/OSLAB/skill/path_resolution.c
/home/monis/OSLAB/skill/pipe_sync.c
/home/monis/OSLAB/skill/pipeline_structure.c
/home/monis/OSLAB/skill/process.c
/home/monis/OSLAB/skill/process_groups.c
/home/monis/OSLAB/skill/process_pipeline.c
/home/monis/OSLAB/skill/resume_jobs.c
/home/monis/OSLAB/skill/shell.c
/home/monis/OSLAB/skill/sigint_forwarding.c
/home/monis/OSLAB/skill/signal_behavior.c

```

Step 2 — Verify your existing reports

ls -l stress_report.txt performance.txt architecture.txt failure_output.txt

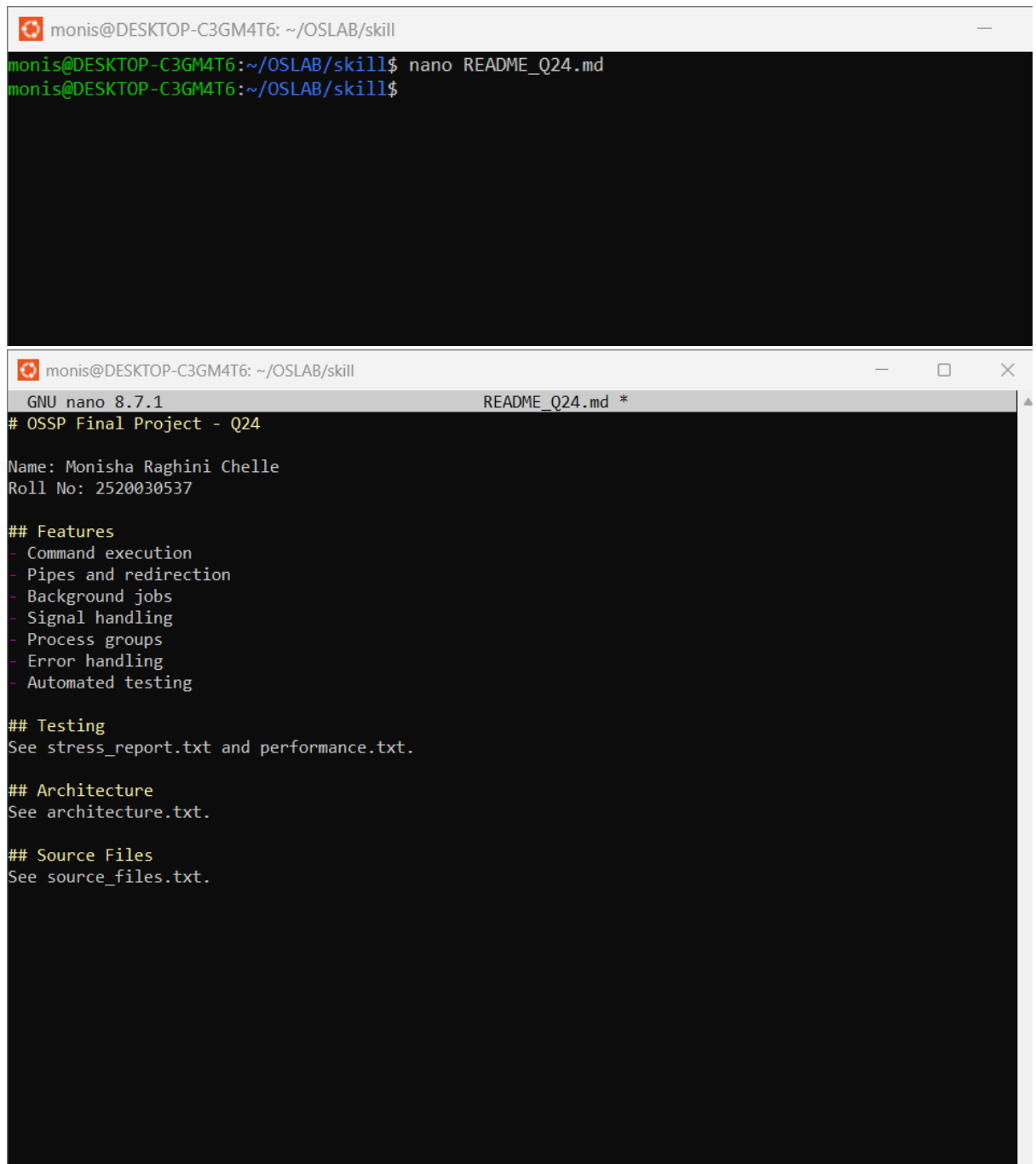
```

monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l stress_report.txt performance.txt architecture.txt failure_output.txt
-rw-r--r-- 1 monis monis 1566 Sep 26 23:40 architecture.txt
-rw-r--r-- 1 monis monis  61 Sep 26 23:38 failure_output.txt
-rw-r--r-- 1 monis monis  746 Sep 26 23:38 performance.txt
-rw-r--r-- 1 monis monis  338 Sep 26 23:38 stress_report.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$

```

Step 3 — Create the Q24 README

nano README_Q24.md



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano README_Q24.md
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$

GNU nano 8.7.1 README_Q24.md *
# OSSP Final Project - Q24

Name: Monisha Raghini Chelle
Roll No: 2520030537

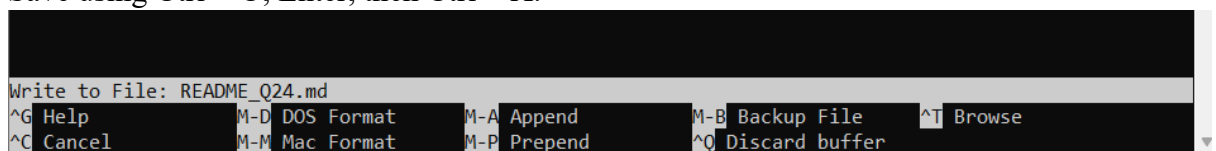
## Features
- Command execution
- Pipes and redirection
- Background jobs
- Signal handling
- Process groups
- Error handling
- Automated testing

## Testing
See stress_report.txt and performance.txt.

## Architecture
See architecture.txt.

## Source Files
See source_files.txt.
```

Save using Ctrl + O, Enter, then Ctrl + X.



```
Write to File: README_Q24.md
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

Step 4 — Final verification

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l source_files.txt README_Q24.md stress_report.txt performance.
txt architecture.txt
-rw-r--r-- 1 monis monis 356 Sep 26 23:55 README_Q24.md
-rw-r--r-- 1 monis monis 1566 Sep 26 23:40 architecture.txt
-rw-r--r-- 1 monis monis 746 Sep 26 23:38 performance.txt
-rw-r--r-- 1 monis monis 2249 Sep 26 23:50 source_files.txt
-rw-r--r-- 1 monis monis 338 Sep 26 23:38 stress_report.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 5 — Create the build script

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
-rw-r--r-- 1 monis monis 338 Sep 26 23:38 stress_report.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ find . -name "integrated_shell.c" -o -name "process_groups.c"
./process_groups.c
./integrated_shell.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano build_Q24.sh
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

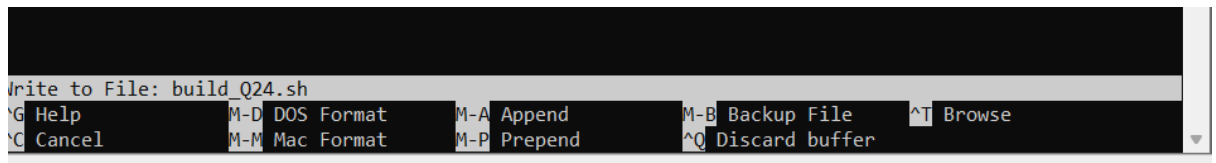
```
GNU nano 8.7.1 build_Q24.sh *
#!/bin/bash
set -e

echo "Q24 - Final Project Build"

gcc -Wall -Wextra integrated_shell.c -o integrated_shell
gcc -Wall -Wextra process_groups.c -o process_groups

echo "Build successful."
```

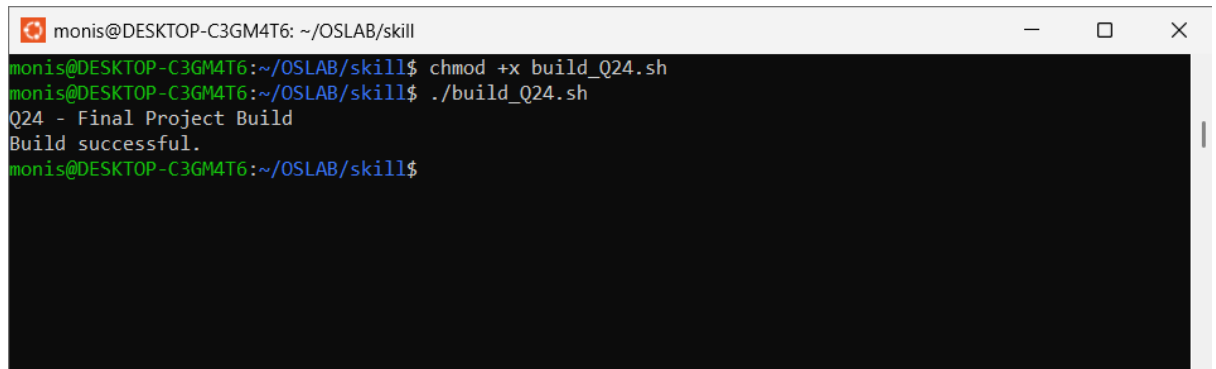
Save using **Ctrl + O**, **Enter**, then **Ctrl + X**.



Run:

chmod +x build_Q24.sh

./build_Q24.sh



Step 6 — Demonstrate the project

A. Command execution and background jobs

./integrated_shell

Enter these commands one at a time:

echo Q24 Final Demo

pwd

sleep 30 &

jobs

exit

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./integrated_shell
Q21 - Integrated Shell
Commands: pwd, cd DIRECTORY, jobs,
          echo TEXT, sleep NUMBER, exit
Add & for background execution.

q21> echo Q24 Final Demo
Q24 Final Demo

q21> pwd
/home/monis/OSLAB/skill

q21> sleep 30 &
[1] Background PID=2503

q21> jobs

ID      PID      COMMAND
1       2503     sleep 30 &

q21> exit
```

B. Pipelines

find ~/OSLAB/skill -name "complex_commands.c"

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ find ~/OSLAB/skill -name "complex_commands.c"
/home/monis/OSLAB/skill/complex_commands.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

C. Signals and terminal control

./process_groups

Enter:

run 60

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./process_groups
Q20 - Process Groups and Terminal Control
Shell PID: 2525
Shell PGID: 2525
Commands: run NUMBER, jobs, fg ID, bg ID, exit

pgshell> run 60

Created job [1].
Child PID: 2526
Child PGID: 2526
Shell PGID: 2525
Press Ctrl+Z to suspend or Ctrl+C to interrupt.
Terminal transferred to PGID 2526.
```

Press Ctrl + Z while the job is running. Then enter:

jobs

fg 1

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./process_groups
Q20 - Process Groups and Terminal Control
Shell PID: 2525
Shell PGID: 2525
Commands: run NUMBER, jobs, fg ID, bg ID, exit

pgshell> run 60

Created job [1].
Child PID: 2526
Child PGID: 2526
Shell PGID: 2525
Press Ctrl+Z to suspend or Ctrl+C to interrupt.
Terminal transferred to PGID 2526.
^] ^] ^] ^] ^Z
Terminal restored to shell PGID 2525.
Job [1] stopped by signal 20.

pgshell> jobs

JOB TABLE
ID      PID      PGID      STATE
1       2526     2526     STOPPED

pgshell> fg 1
Terminal transferred to PGID 2526.
Foreground job resumed.
```

The stopped job should resume in the foreground. Press Ctrl + C to demonstrate interrupt handling, then enter exit.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./process_groups
Q20 - Process Groups and Terminal Control
Shell PID: 2525
Shell PGID: 2525
Commands: run NUMBER, jobs, fg ID, bg ID, exit

pgshell> run 60

Created job [1].
Child PID: 2526
Child PGID: 2526
Shell PGID: 2525
Press Ctrl+Z to suspend or Ctrl+C to interrupt.
Terminal transferred to PGID 2526.
^]^^^]^^^Z
Terminal restored to shell PGID 2525.
Job [1] stopped by signal 20.

pgshell> jobs

JOB TABLE
ID      PID      PGID      STATE
1       2526     2526     STOPPED

pgshell> fg 1
Terminal transferred to PGID 2526.
Foreground job resumed.
^]^^C
Terminal restored to shell PGID 2525.
Job [1] terminated by signal 2.

pgshell> exit
Process group program exited.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 7 — Final verification

```
ls -l README_Q24.md source_files.txt build_Q24.sh
cat stress_report.txt
cat architecture.txt
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l README_Q24.md source_files.txt build_Q24.sh
-rw-r--r-- 1 monis monis 356 Sep 26 23:55 README_Q24.md
-rwxr-xr-x 1 monis monis 190 Sep 26 23:58 build_Q24.sh
-rw-r--r-- 1 monis monis 2249 Sep 26 23:50 source_files.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat stress_report.txt
Q23 - STRESS TEST REPORT
Date: Sat Sep 26 23:38:20 IST 2026
=====
TEST 1: Large pipeline
Pipeline: PASS
TEST 2: Multiple background jobs
Launched 5 jobs
Background jobs finished.
TEST 3: Performance and resources
Performance test: PASS
TEST 4: Error detection
Error detection: PASS
Total failures: 0
FINAL RESULT: PASS
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat architecture.txt
Q23 - SHELL ARCHITECTURE DOCUMENTATION

1. ARCHITECTURE

User Input
|
v
Command Parser
|
v
Syntax Validation
|
v
Built-in Command? ---- YES ---> Execute in Shell
|
NO
|
v
Create Child Process
|
v
Execute Command
|
+---- Foreground ---> Wait for Completion
|
+---- Background ---> Store Job Information
|
v
Monitor Job Status
```

```
+---- Foreground ---> Wait for Completion
|
+---- Background ---> Store Job Information
|
v
Monitor Job Status
```

2. EXECUTION PATHS

Foreground:

Read -> Parse -> fork -> exec -> waitpid -> Prompt

Background:

Read -> Parse -> fork -> exec -> Store PID -> Prompt

Pipeline:

Read -> Parse -> pipe -> fork children ->

dup2 -> exec -> close unused descriptors -> waitpid

3. IMPORTANT SYSTEM CALLS

fork()	Creates a child process.
execvp()	Executes a command.
pipe()	Creates a communication pipe.
dup2()	Redirects input or output.
waitpid()	Waits for or monitors child processes.
setpgid()	Assigns a process group.
tcsetpgrp()	Transfers terminal control.
kill()	Sends signals to processes or groups.

4. DESIGN DECISIONS

- Separate parsing from command execution.
- Execute external commands in child processes.
- Use process groups for job control.
- Track background jobs using process IDs.
- Report errors without terminating the shell.
- Close unused file descriptors.
- Release resources after execution.

5. TESTING

Large pipelines: See stress_report.txt

Multiple jobs: See stress_report.txt

Performance: See performance.txt

Error detection: See failure_output.txt

monis@DESKTOP-C3GM4T6:~/OSLAB/skill\$